

5.2.14

EE25BTECH11011 - Navya

Question:

Solve the following system of linear equations.

$$x - 3y - 3 = 0 \quad (1)$$

$$3x - 9y - 2 = 0 \quad (2)$$

Solution:

The given equations of lines are,

$$(1 \quad -3)\mathbf{x} = 3 \quad (3)$$

$$(3 \quad -9)\mathbf{x} = 2 \quad (4)$$

In matrix form,

$$\begin{pmatrix} 1 & -3 \\ 3 & -9 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (5)$$

So the augmented matrix is

$$\left(\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & -9 & 2 \end{array} \right) \quad (6)$$

Performing row operations:

$$R_2 \rightarrow R_2 - 3R_1 \implies \left(\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 0 & -7 \end{array} \right) \quad (7)$$

This corresponds to the equation

$$0x + 0y = -7, \quad (8)$$

which is a contradiction.

Hence, the system has **no solution**. The two equations can be written as

$$x - 3y = 3, \quad x - 3y = \frac{2}{3}, \quad (9)$$

which represent two parallel lines with the same slope but different intercepts.

Therefore, The given system of equations is **inconsistent**, and the two lines are parallel.

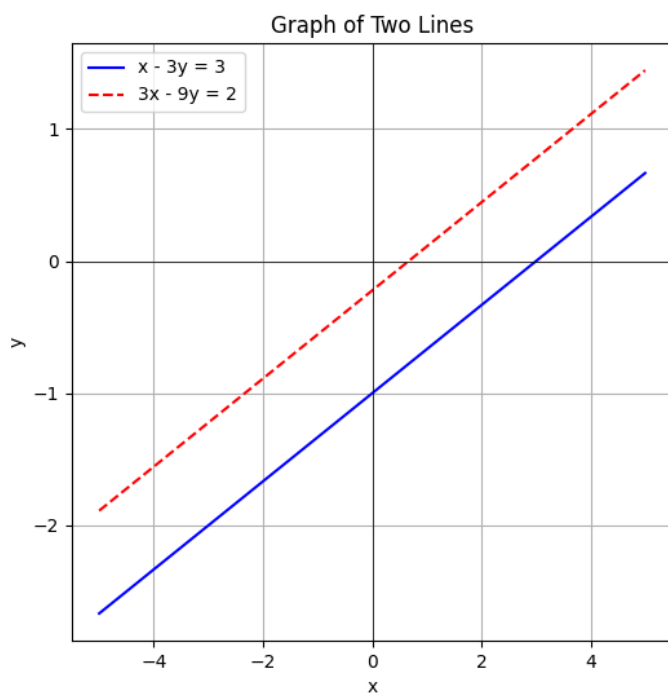


Fig. 0: Parallel lines